

PE loss: Perception-enhanced distortion-oriented loss for image restoration

Meng Li¹ (✉), Yuhao Wang², Yu Zhong¹, Xishan Zhang¹, and Shaoli Liu¹

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Abstract Image restoration is the inverse problem of recovering high-quality images from knowledge of degraded images; it includes image super-resolution, image denoising, image deblurring, etc. The objective of image restoration methods is to minimize the error defined by the loss function between the network output image and the corresponding ground truth. Distortion-oriented loss functions are fundamental and commonly used in image restoration. However, these functions treat all pixels as equally important without differentiating between sharp and blurred edge areas, which does not match human visual perception. As a result, these methods can produce accurate but blurred images. To address this issue and achieve both accurate and perceptually satisfactory results, we propose a novel perception-enhanced distortion-oriented loss (PE loss) for image restoration, inspired by the Mach band effect. This effect demonstrates that sharp edges are perceived as having better quality than blurred edges by the human visual system. Our approach includes designing a blur factor map that detects blurred pixels and penalizes them by amplifying their error. The PE loss is a simple yet effective plug-and-play method, and we apply it to state-of-the-art networks. Extensive quantitative and qualitative experiments show that our method can restore images with sharp edges and high perceptual quality.

Keywords image restoration; perceptual loss; super-resolution; denoising; convolutional neural networks

1 Introduction

Image restoration aims to recover a high-quality image from a degraded one. This ill-posed problem has remained challenging for decades. In recent years, neural network (NN)-based methods have achieved significant success in image restoration. Some research focuses on well-designed network architectures, while others emphasize the loss function as the optimization objective.

NN-based methods for image restoration typically use degraded images as input and train the model by minimizing the error, as defined by a loss function, between the output image and the corresponding ground truth. Commonly used loss functions can be classified into perception-oriented losses and distortion-oriented losses. Perception-oriented losses compare the difference between output and ground truth images in the feature space. Examples include perceptual loss [1, 2] and adversarial loss [3–5]. While these methods achieve good perceptual quality, they can sometimes produce unnatural artifacts alongside the reconstructed textures. In practical applications, perception-oriented losses are often combined with distortion-oriented losses to reduce visual artifacts [6–8]. Distortion-oriented losses minimize the distance between the output pixel values and the ground truth pixel values. Examples include mean-squared error (MSE or L_2) [9–13] and mean-absolute error (MAE or L_1) [14, 15]. While these methods typically produce high peak signal-to-noise ratio (PSNR) values, they can sometimes result in blurred output.

Distortion-oriented loss functions, such as L_1 loss, are based on mathematical measurements of similarity but do not match human visual perception. These loss functions treat all pixels as equally important, failing to differentiate between blurred and sharp edge

1 Intelligent Processor Research Center, Institute of Computing Technology, Chinese Academy of Sciences, Beijing 100190, China. E-mail: M. Li, limeng2020@ict.ac.cn (✉); Y. Zhong, zhongyu21s@ict.ac.cn; X. Zhang, zhangxishan@ict.ac.cn; S. Liu, liushaoli@ict.ac.cn.

2 School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China. E-mail: ustcwylh@mail.ustc.edu.cn.

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areas. However, sharp edges are perceived as having better quality than blurred edges due to the Mach band effect [16, 17], an inherent edge enhancement mechanism of the retina. As a result, absolute pixel error does not effectively represent perceptual error between images without accounting for the differences between blurred and sharp edge areas.

To obtain natural, perceptually realistic images, we propose a novel perception-enhanced distortion-oriented loss for image restoration, which is based on perceptual differences between blurred and sharp edges as perceived by the human visual system (HVS). In this approach, blurred pixels in the degraded image are penalized more heavily than sharp pixels due to their poorer perceptual quality. Our algorithm consists of two stages. Firstly, we use a blur factor map to identify blurred pixels in the degraded images. Specifically, by analyzing the second-order gradient map of the high-quality image and combining it with the pixel differences between the degraded image and the high-quality image, we can effectively detect blurred pixels. Secondly, we penalize blurred pixels by assigning them a higher weight in the error calculation during network optimization. This approach enhances preservation of image structures and improves perceptual quality compared to traditional distortion-oriented loss functions, as demonstrated in Fig. 1.

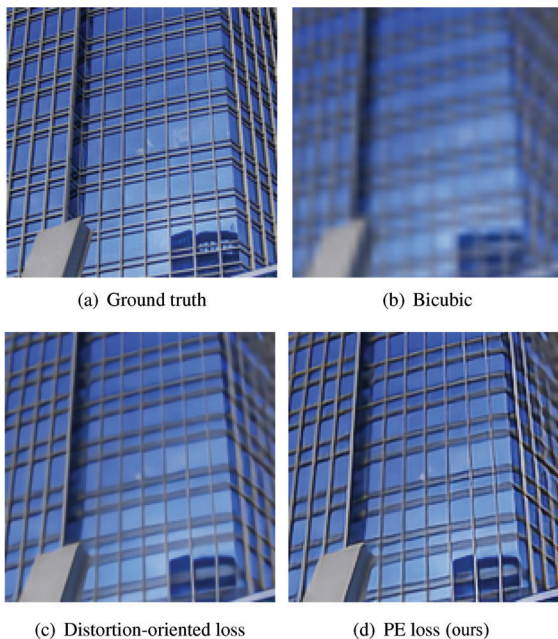


Fig. 1 4× super-resolution result using a model optimized by L_1 loss compared to the same model optimized by our PE loss. Our result has better perceptual quality with sharper edges.

We evaluate the effectiveness of the perception-enhanced distortion oriented loss (PE loss) through extensive experiments incorporating it into popular image restoration networks. These include networks for image super-resolution [18], image denoising [19–22], and image deblurring [23–26]. We apply our PE loss to retrain these networks and compare the results to those obtained using the original networks. Both quantitative and qualitative experimental results demonstrate that the PE loss effectively enhances the sharpness and fidelity of the restored images compared to the original distortion-oriented or perception-oriented loss functions.

The main contributions of this paper are as follows:

- We propose a novel PE loss for image restoration tasks. We detect and represent blurred pixels in a map, which is used to penalize them more heavily by amplifying their errors, thereby achieving improved perceptual quality.
- We introduce the PE loss in a plug-and-play manner, incorporate it into state-of-the-art networks and illustrate the improvement on both distortion-oriented loss and perception-oriented loss methods.
- We demonstrate the effectiveness of the PE loss through extensive quantitative and qualitative experiments on popular image restoration networks. Our evaluation covers image super-resolution, image denoising, and image deblurring tasks.

2 Methodology

2.1 Motivation

The human visual system (HVS) is highly sensitive to edges in images. The Mach band effect demonstrates that the HVS automatically enhances edges in images, reflecting an intrinsic edge enhancement mechanism of the retina, as shown in Fig. 2. Consequently, sharp edges are more visually acceptable to the human eye, which provides valuable insight for designing loss functions in image restoration tasks. As Fig. 3 shows, sharp edges are perceptually closer to natural edges than blurred ones.

In the image super-resolution task, a high-resolution image is downsampled to a lower resolution by removing pixels or using interpolation methods such as bilinear, bicubic, or nearest-neighbor. This

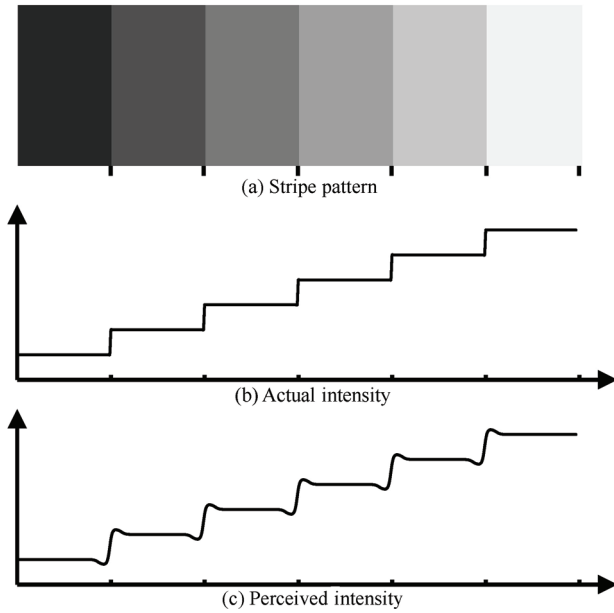


Fig. 2 Mach band effect. Ernst Mach described the phenomenon that the HVS tends to undershoot or overshoot around the boundary of regions of different intensities. Although the intensity of the stripes is constant, we actually perceive a brightness pattern that is strongly scalloped near the boundaries.

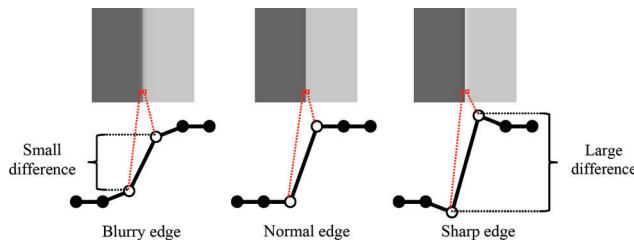


Fig. 3 Blurred and sharp edges.

process leads to the loss of image details, edges, and textures, and also introduces slight blurring. In the image denoising task, the primary degradation occurs through the addition of noise to a clean image, such as Gaussian noise, salt-and-pepper noise, Poisson noise, etc. However, the boundary between noise and image texture is very subtle, which can lead to the removal of certain original image details during denoising, causing blurring. In the image deblurring task, the main degradation involves the introduction of blurring effects, such as motion blur caused by movement during exposure, defocusing blur from out-of-focus conditions, and other factors leading to synthetic degradation, resulting in a loss of sharpness and edge details.

Without loss of generality, we explain our approach using the super-resolution task. In this context, we denote the high-resolution image, low-resolution

image, and reconstructed image as HR, LR, and SR, respectively.

Distortion-oriented losses, which are widely used loss functions, are perception-agnostic and do not account for perceptual differences between the output and ground-truth images. The value of a distortion-oriented loss depends solely on the absolute errors of pixels in the SR image compared to the HR image. As the green line in Fig. 4 shows, when the absolute value of the pixel difference between the SR image and the HR image is the same, the loss remains the same, regardless of whether the pixel is blurred or sharp. However, sharp images are generally more acceptable to the HVS than blurred ones.

Pixel error alone is insufficient to reflect the visual quality of an image. Since sharp images generally have better perceptual quality than blurred ones, the loss associated with blurred pixels in the SR image should be greater than that for sharp pixels, even if their pixel errors are the same. As the red line in Fig. 4 shows, this approach better matches similarity perceived by the HVS. Inspired by this observation, we propose the perception-enhanced distortion-oriented loss (PE loss).

2.2 Perception-enhanced distortion-oriented loss (PE loss)

PE loss is a perception-aware distortion-oriented loss in which weights of blurred pixel errors are greater than for other pixels. Firstly, we compute a blur factor map to detect blurred pixels in the SR image.

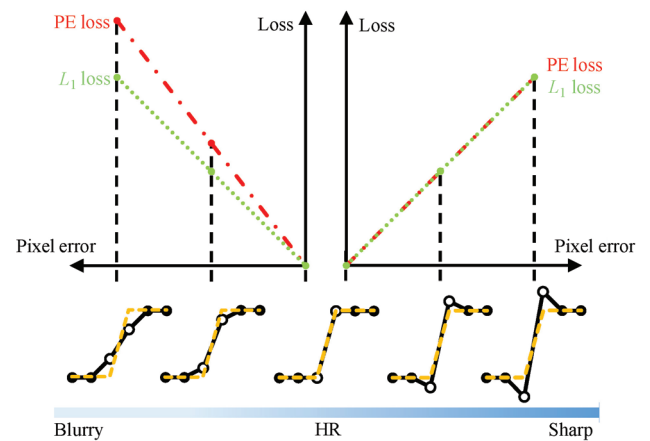


Fig. 4 Comparison of L_1 and PE losses. The left figure shows losses for blurred SR edges. The right figure shows losses for sharp SR edges. The black curves below show different SR edges, while the yellow ones indicate specific edges in natural images. PE loss penalizes blurred edges more than L_1 loss. Both losses have the same value for sharp edges.

Secondly, we calculate the weight of each pixel error between the SR image and the HR image separately. Finally, we take the weighted average of pixel errors as the loss. The whole pipeline is shown in Fig. 5.

2.2.1 Detecting blurred pixels

Blurred pixels can be detected by combining the second-order gradient of the HR image with the difference between the SR image and the HR image. For edges, the pixel difference on both sides of the edge in the blurred image is smaller than that in the sharp image (see Fig. 3). The second-order gradient reflects the direction and intensity of changes in image pixels. By combining these two properties, blurred pixels can be detected. The underlying principle is explained below.

Without loss of generality, we assume that the edge we are analyzing is vertical. In the original HR image, the intensity $f(x, y)$ at position (x, y) can be expressed as

$$f(x, y) = s(x, y)I_1 + (1 - s(x, y))I_2 \quad (1)$$

where I_1 and I_2 denote the intensity on either side of the edge, respectively. $s(x, y)$ is a sign function that describes an edge at $x = 0$, and can be expressed as

$$s(x, y) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (2)$$

The second-order gradient of $f(x, y)$ is

$$\begin{aligned} \nabla^2 f &= \nabla^2 s(I_1 - I_2) \\ &= \begin{cases} I_1 - I_2, & x = -1 \\ I_2 - I_1, & x = 0 \\ 0, & x \neq 0 \end{cases} \end{aligned} \quad (3)$$

Let $f(0, y)$, $f_b(0, y)$, and $f_s(0, y)$ denote the intensity of the lines at $x = 0$ in the HR image, the blurred SR image, and the sharp SR image. Without loss of generality, we assume that $I_1 > I_2$. The value of $f_b(0, y)$ is I_b ($I_2 < I_b < I_1$). The value of $f_s(0, y)$ is I_s ($I_s > I_1$). The pixel difference between the blurred line and the original line is

$$f_b(0, y) - f(0, y) = I_b - I_1 < 0 \quad (4)$$

and the pixel difference between the sharp line and the original line is

$$f_s(0, y) - f(0, y) = I_s - I_1 > 0 \quad (5)$$

The second-order gradient of this line in the HR image is

$$\nabla^2 f(0, y) = I_2 - I_1 < 0 \quad (6)$$

Combining Eqs. (6) and (4), if the pixels in the SR image are blurred, the sign of the pixel difference is the same as the sign of the second gradient of the HR image. Combining Eqs. (6) and (5), if the pixels are sharp, the sign of the pixel difference has opposite sign to the second-order gradient of the HR image.

Thus, blurred pixels can be detected by comparing the sign of the pixel difference and the sign of the second-order gradient. We use this principle to design our blur factor map. We first make the following definitions. $p_{i,j}^{\text{SR}}$ and $p_{i,j}^{\text{HR}}$ are the intensities of the pixel at position (i, j) in the SR image I^{SR} and the HR image I^{HR} , respectively. $g_{i,j}$ is the value of the second-order gradient $\nabla^2 I^{\text{HR}}$ at position (i, j) , calculated by applying the Laplace operator to the HR image. The

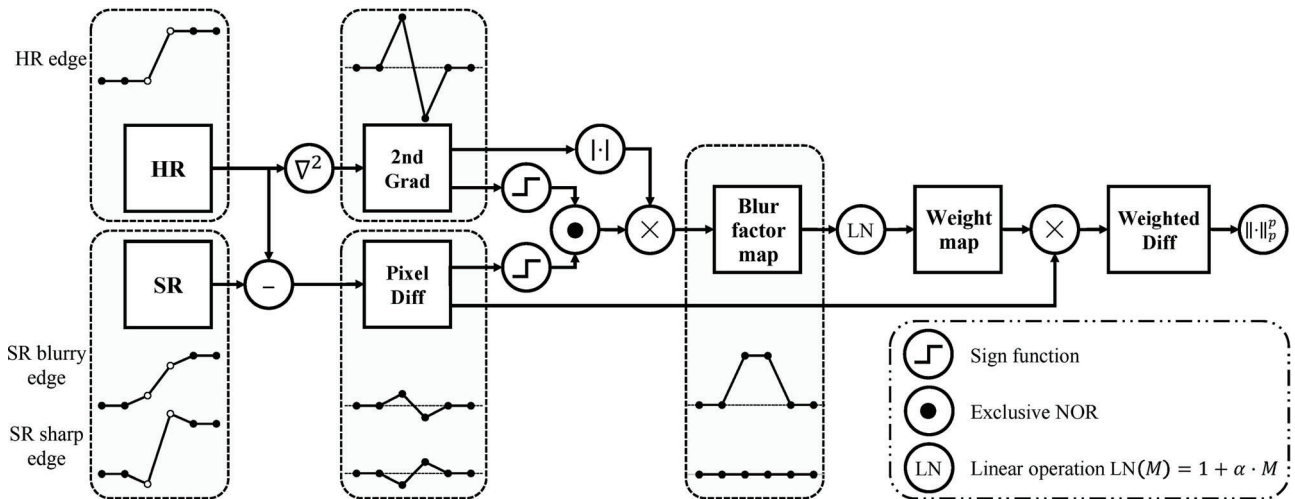


Fig. 5 Pipeline of PE loss. The two key steps in computing PE loss are detecting blurred pixels and penalizing blurred pixels. The blurred pixels are detected by the *blur factor map*, which is calculated using pixel differences and the second-order gradient, as defined as Eq. (8). The weight for the pixel difference is given by the blur factor map, as defined as Eq. (9). Blurred pixels are penalized by increasing the weight of pixel differences in the PE loss.

difference between the SR image and the HR image at position (i, j) is defined as

$$d_{i,j} = I_{i,j}^{\text{SR}} - I_{i,j}^{\text{HR}} \quad (7)$$

The blur factor map has the same size as the image; if a pixel is blurred, the corresponding value in the blur factor map is the absolute value of the second-order gradient in the HR image. A value of 0 means a typical pixel. Thus, the blur factor map $M = \{m_{i,j}\}$, where $m_{i,j}$ at position (i, j) is defined as

$$m_{i,j} = \begin{cases} |g_{i,j}|, & g_{i,j} > 0, d_{i,j} > 0 \\ 0, & g_{i,j} > 0, d_{i,j} \leq 0 \\ 0, & g_{i,j} \leq 0, d_{i,j} > 0 \\ |g_{i,j}|, & g_{i,j} \leq 0, d_{i,j} \leq 0 \end{cases} \quad (8)$$

2.2.2 Penalizing blurred pixels

In our PE loss, the weights applied to the pixel differences between the SR image and the HR image is determined by the blur factor map. The basic weight for differences is 1; blurred pixels in the SR image are given greater weights. The absolute value of the second-order gradient indicates the strength of the edge in the area. For obvious edges in the HR image, the negative effect of blurring is greater. Therefore, for blurred pixels, the weight of the difference is adjusted according to the absolute value of the second gradient of the HR image. The weight W is defined as

$$W = 1 + \alpha M \quad (9)$$

where α denotes the penalty factor applied to blurred pixels.

Finally, the PE loss $\mathcal{L}^{\text{PE}}$ is defined as the L_p norm of the weighted pixel difference as Eq. (10):

$$\mathcal{L}^{\text{PE}} = \mathbb{E}_{I^{\text{SR}}} \|W(I^{\text{SR}} - I^{\text{HR}})\|_p^p \quad (10)$$

where $\|\cdot\|_p$ denotes the L_p norm. The process of training the backbone network through the PE loss is shown in Fig. 6.

PE loss can be used independently instead of distortion-oriented losses or together with other perception-oriented losses to further improve perceptual quality.

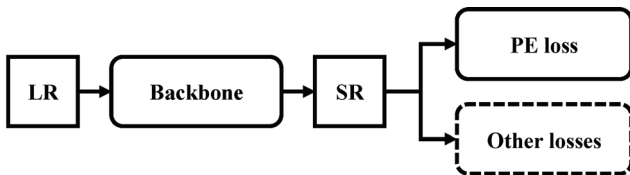


Fig. 6 Pipeline for training the backbone network using PE loss. PE loss can be used as the only loss function, or with other perception-driven loss functions.

3 Experiments

3.1 Approach

In this section, we introduce the training method and evaluation metrics, discuss the experimental setup and study the impact of penalty factor α . We evaluate the effectiveness of the proposed method on state-of-the-art backbone network architectures for different image restoration tasks, including image super-resolution, denoising, and deblurring. Our experiments show that the proposed PE loss is beneficial to neural networks with both distortion-oriented losses and perception-oriented losses.

3.2 Experimental setting

3.2.1 Training

We conducted experiments on image super-resolution, image denoising, and image deblurring tasks, including distortion-oriented methods and perception-oriented methods. We retrained each state-of-the-art backbone network architecture by replacing the distortion-oriented loss with PE loss and compared our results to the results of the original methods. For distortion-oriented methods, we used PE loss as the only loss function to retrain the networks. For perception-oriented methods, we use PE loss together with the original perception-oriented losses to retrain the networks. For a fair comparison, we train the network architectures using the same parameters as for the original methods. To match each original method, the L_p norm of PE loss was the same as in the original method. All experiments were implemented in PyTorch using NVIDIA Tesla A100 GPUs. Further training details and reproduction methods are given in each subsection.

3.2.2 Evaluation metrics

We chose four commonly used metrics for all tasks: PSNR, structural similarity (SSIM) [29], learned perceptual image patch similarity (LPIPS), and natural image quality evaluator (NIQE) [30]. Lower LPIPS and NIQE values indicate higher perceptual quality. Note that we evaluated the results using PSNR, SSIM, and NIQE on the Y channel in YCbCr space.

3.2.3 Penalty factor

In PE loss, the penalty factor α is a crucial parameter that influences image quality and metrics. To explore the effect of the α parameter, we conducted

studies varying the α value for typical networks for three different tasks: super-resolution, denoising, and deblurring. For the image super-resolution task, we used the NLSN network architecture. Each metric represents average results across four test sets: Set5, Set14, B100 [31], and Urban100 [32]. For the image denoising task, the backbone network architecture was NBNNet. Metrics here were averaged across three test sets: DIV2K [33], KODAK24 [34], and CBSD68 [35]. For the image deblurring task, we employed the Restormer network architecture. Metrics for this task were averaged over three test sets: GoPro, HIDE [36], and RealBlur [37]. We trained each of these three backbone networks using PE loss with various values of α : 1.0, 2.0, 3.0, 4.0, 6.0, 8.0, and 10.0. All other experimental settings remained constant.

We used the original results trained with L_1 loss as a reference; Table 1 presents the results of models trained with PE loss for various α values. The best result in each row is highlighted. The experiments indicate that choosing an appropriate value of α can enhance network performance and yield better results.

In subsequent experiments, to alleviate concerns

that the performance gains might be due to over-tuned parameters, we compared results across different tasks using a consistent value of $\alpha = 2.0$.

3.3 PE loss and image super-resolution

3.3.1 Backbone network architectures

We tested PE loss with distortion-oriented methods EDSR, RRDBNet, NLSN, and ATD, and perception-oriented methods SRGAN, ESRGAN, and SPSR, with a scaling factor of $\times 4$ between LR and HR images. The original loss function used by EDSR, RRDBNet, NLSN, and ATD is L_1 loss. The original loss functions used by SRGAN are composed of MSE loss, perceptual loss and adversarial loss. The original loss functions of ESRGAN and SPSR were composed of L_1 loss, perceptual loss and adversarial loss. The models trained using PE loss are denoted EDSR-PE, RRDBNet-PE, NLSN-PE, ATD-PE, SRGAN-PE, ESRGAN-PE, and SPSR-PE, respectively.

3.3.2 Datasets

Consistent with the original method, we used DIV2K as the training dataset for EDSR-PE, NLSN-PE, SRGAN-PE, and SPSR-PE. We used DIV2K and Flickr2K [38] as training datasets for RRDBNet-PE,

Table 1 Experimental results for different values of penalty factor α , for different tasks. The backbone network for image super resolution was NLSN. Each metric represents average results on four test sets: Set5, Set14, B100, and Urban100. The backbone network for image denoising was NBNNet. Each metric represents an average of results on three test sets: DIV2K, KODAK24, and CBSD68. The backbone network for image deblurring was Restormer. Each metric represents average results on three test sets: GoPro, HIDE, and RealBlur. Best results are highlighted in bold

Task	Metric	$L_1(\alpha = 0.0)$	$\alpha = 1.0$	$\alpha = 2.0$	$\alpha = 3.0$	$\alpha = 4.0$	$\alpha = 6.0$	$\alpha = 8.0$	$\alpha = 10.0$
Super resolution	LPIPS $\downarrow$	0.2479	0.2397	0.2357	0.2335	0.2324	0.2315	0.2314	0.2315
	NIQE $\downarrow$	6.2195	6.1075	6.0376	5.9905	5.9531	5.9058	5.8582	5.8179
	SSIM $\uparrow$	0.8113	0.8133	0.8133	0.8128	0.8121	0.8104	0.8086	0.8071
	PSNR $\uparrow$	29.14	29.00	28.81	28.63	28.47	28.19	27.97	27.79
Denoise- $\sigma=25$	LPIPS $\downarrow$	0.0814	0.0723	0.0717	0.0722	0.0722	0.0740	0.0782	0.0810
	NIQE $\downarrow$	3.9496	3.8316	3.8105	3.9275	3.9383	4.0467	4.1717	4.3057
	SSIM $\uparrow$	0.8760	0.8787	0.8778	0.8758	0.8738	0.8683	0.8637	0.8593
	PSNR $\uparrow$	31.49	31.39	31.12	30.86	30.63	30.15	29.78	29.52
Denoise- $\sigma=50$	LPIPS $\downarrow$	0.1553	0.1406	0.1375	0.1374	0.1373	0.1417	0.1467	0.1511
	NIQE $\downarrow$	4.5187	4.3948	4.3189	4.3861	4.3761	4.3918	4.4646	4.5289
	SSIM $\uparrow$	0.7824	0.7872	0.7861	0.7840	0.7818	0.7742	0.7690	0.7636
	PSNR $\uparrow$	27.16	27.13	26.96	26.79	26.64	26.30	26.06	25.86
Denoise- $\sigma=75$	LPIPS $\downarrow$	0.2168	0.1990	0.1940	0.1929	0.1925	0.1983	0.2032	0.2081
	NIQE $\downarrow$	4.8904	4.8601	4.8129	4.7789	4.7593	4.7786	4.8375	4.8497
	SSIM $\uparrow$	0.7100	0.7158	0.7155	0.7141	0.7121	0.7045	0.7003	0.6951
	PSNR $\uparrow$	24.01	24.02	23.94	23.85	23.78	23.57	23.44	23.31
Deblur	LPIPS $\downarrow$	0.0754	0.0743	0.0684	0.0710	0.0707	0.0707	0.0709	0.0717
	NIQE $\downarrow$	5.7795	5.6466	5.5488	5.4753	5.4227	5.3431	5.2866	5.2523
	SSIM $\uparrow$	0.9095	0.9033	0.9028	0.9021	0.9011	0.8996	0.8978	0.8961
	PSNR $\uparrow$	31.19	30.87	30.77	30.66	30.55	30.37	30.22	30.08

ESRGAN-PE, and ATD-PE. We used four commonly used benchmarks for all testing: Set5, Set14, B100, and Urban100.

3.3.3 Quantitative comparison

A quantitative comparison of results for distortion-oriented methods is shown in Table 2. For each pair of results, the better result is highlighted. We can see that all models with PE loss achieve better LPIPS and NIQE than the original models: the models optimized by PE loss produce results with better perceptual quality. Especially on the ATD network,

nearly all metrics show improvement. Except for the Set5 testing dataset, models with PE loss achieve better SSIM than the original methods. On average, the SSIM values of the models optimized by PE loss are higher, which means that the models optimized by PE loss better preserve the image structures. Since L_1 loss directly optimizes PSNR, methods only using L_1 loss obtain higher PSNR but lower perceptual quality.

A quantitative comparison of results for perception-oriented methods is shown in Table 3. Since Set5 has

Table 2 Quantitative results on benchmark datasets for $4\times$ distortion-oriented image super-resolution methods. Each block compares results of each model trained with PE loss and the original method. Better results are highlighted in bold. On average, models with PE loss achieve better SSIM, LPIPS, and NIQE. Since L_1 loss directly optimizes PSNR, methods trained using L_1 loss obtain higher PSNR

Dataset	Metric	EDSR [27]	EDSR-PE	RRDBNet	RRDBNet-PE	NLSN [28]	NLSN-PE	ATD	ATD-PE
Set5	LPIPS ↓	0.1861	0.1785	0.1843	0.1799	0.1684	0.1649	0.1798	0.1762
	NIQE ↓	7.1441	6.8699	7.1230	6.9179	7.0472	6.9121	7.0553	6.9245
	SSIM ↑	0.8988	0.8982	0.9011	0.9003	0.9000	0.8999	0.9042	0.9047
	PSNR ↑	32.48	32.11	32.73	32.39	32.70	32.35	32.92	32.93
Set14	LPIPS ↓	0.2782	0.2667	0.2757	0.2642	0.2716	0.2626	0.2743	0.2707
	NIQE ↓	6.1305	6.0522	6.1375	5.9915	6.0856	5.8453	6.2453	6.2456
	SSIM ↑	0.7879	0.7879	0.7915	0.7926	0.7889	0.7904	0.7943	0.7959
	PSNR ↑	28.81	28.33	29.00	28.58	28.95	28.56	29.13	29.14
B100	LPIPS ↓	0.3763	0.3520	0.3723	0.3502	0.3568	0.3377	0.3606	0.3563
	NIQE ↓	6.3498	5.9326	6.3598	6.0498	6.2978	5.9825	6.1823	6.1343
	SSIM ↑	0.7420	0.7454	0.7453	0.7491	0.7416	0.7463	0.7495	0.7509
	PSNR ↑	27.72	27.41	27.84	27.52	27.81	27.56	27.91	27.92
Urban100	LPIPS ↓	0.2083	0.1877	0.2005	0.1804	0.1949	0.1774	0.1824	0.1768
	NIQE ↓	5.4342	5.4191	5.6235	5.6156	5.4475	5.4104	5.4264	5.3803
	SSIM ↑	0.8036	0.8044	0.8152	0.8157	0.8146	0.8166	0.8286	0.8324
	PSNR ↑	26.65	26.22	27.03	26.53	27.11	26.77	27.62	27.74

Table 3 Quantitative results on benchmark datasets for $4\times$ perception-oriented image super-resolution methods. Each block compares results of each model trained with PE loss and the original method. Better results are highlighted in bold. On average, the PSNR, SSIM, LPIPS, and NIQE metrics can be improved by replacing L_1 loss by PE loss for perception-oriented image super-resolution methods

Dataset	Metric	SRGAN	SRGAN-PE	ESRGAN	ESRGAN-PE	SPSR	SPSR-PE
Set5	LPIPS ↓	0.0795	0.0818	0.0806	0.0714	0.0694	0.0690
	NIQE ↓	4.8278	4.9064	5.2280	5.0842	4.6770	4.9263
	SSIM ↑	0.8469	0.8524	0.8516	0.8460	0.8432	0.8488
	PSNR ↑	29.93	30.19	30.46	30.44	30.39	30.61
Set14	LPIPS ↓	0.1484	0.1443	0.1374	0.1271	0.1353	0.1316
	NIQE ↓	3.7865	3.6143	3.8774	3.9189	3.8489	3.8003
	SSIM ↑	0.7075	0.7195	0.6980	0.7087	0.7133	0.7164
	PSNR ↑	26.53	26.85	26.28	26.66	26.65	26.75
B100	LPIPS ↓	0.1463	0.1434	0.1695	0.1691	0.1693	0.1662
	NIQE ↓	3.6534	3.6452	3.6636	3.4855	3.3867	3.3260
	SSIM ↑	0.7298	0.7377	0.6495	0.6613	0.6571	0.6637
	PSNR ↑	24.41	24.64	25.29	25.59	25.49	25.70
Urban100	LPIPS ↓	0.1861	0.1862	0.1261	0.1222	0.1212	0.1202
	NIQE ↓	3.3174	3.2083	4.2066	3.8883	3.7891	3.7509
	SSIM ↑	0.6542	0.6613	0.7327	0.7476	0.7463	0.7465
	PSNR ↑	25.55	25.75	24.35	24.82	24.79	24.84

only 5 images, the results deviate from those for other test sets. For other test sets, all models with PE loss achieve better SSIM, LPIPS, and NIQE than the original models, as for distortion-oriented methods. In addition, the PSNR values of our results are also higher than for the original models for most test sets. Typically, PSNR, SSIM, LPIPS, and NIQE can be improved by replacing the L_1 loss by PE loss for the perception-oriented methods. The quantitative results show that in comparison to the original methods, PE loss can not only improve the accuracy of SR images but also better preserve the structure and bring better perceptual quality.

3.3.4 Qualitative comparison

We also compare visual effects for distortion-oriented methods and perception-oriented methods. Figure 7

shows a visual comparison of results for state-of-the-art distortion-oriented methods. In the first image, the details of our result are easier to distinguish than the blurred original result: our PE loss preserves the structures in images better. Our results are sharper than the original result for the second image, and have better perceptual quality. In other rows, our method also recovers better textures and sharper edges than the original methods. These qualitative results show that compared to distortion-oriented loss, models optimized by PE loss can restore correct images while preserving the image structure.

Figure 8 shows a visual comparison of results for state-of-the-art perception-oriented methods. It shows that our model recovers sharper edges with fewer artifacts than the original models. The PE loss can

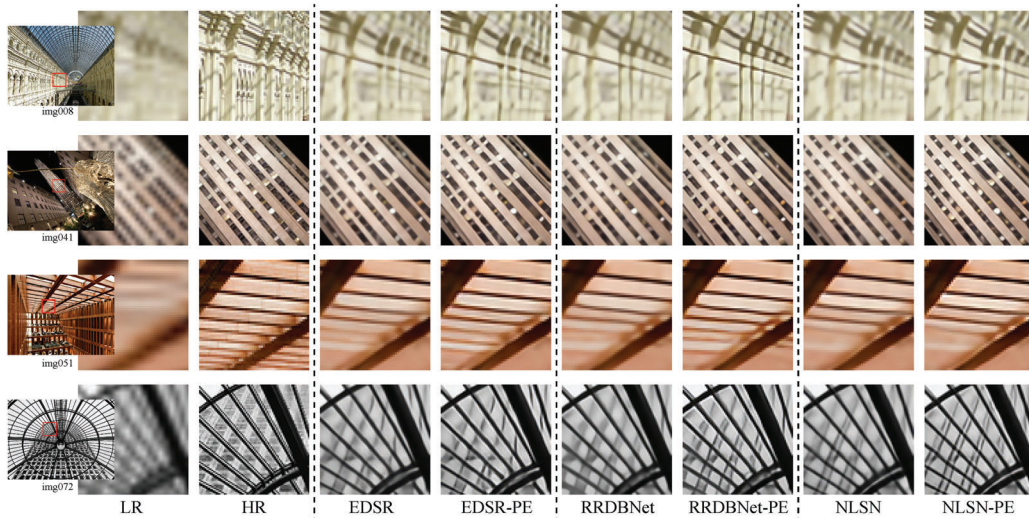


Fig. 7 Visual comparison of distortion-oriented methods for 4× super-resolution on Urban100 dataset.

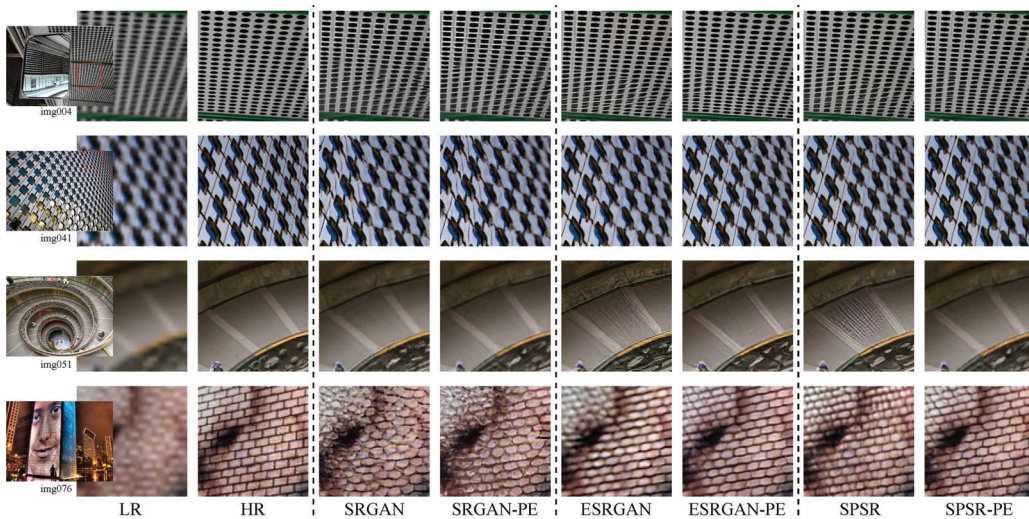


Fig. 8 Visual comparison of perception-oriented methods for 4× super-resolution on Urban100 dataset.

correctly locate the edge, which helps the models generate more realistic SR images without severe distortions when used together with perception-oriented losses.

3.4 PE loss and image denoising

3.4.1 Backbone network architectures

We also demonstrate the effectiveness of PE loss in image denoising. We evaluated PE loss with state-of-the-art network architectures FFDNet, RIDNet, SADNet, NBNet, and CTNet. The original loss function for all these networks was L_1 loss. The models trained with PE loss are denoted FFDNet-PE, RIDNet-PE, SADNet-PE, NBNet-PE, and CTNet-PE, respectively.

3.4.2 Datasets

We adopted DIV2K which contains 800 images with 2K resolution as the training dataset and added Gaussian noise with different intensities ($\sigma = 25, 50, 75$) to synthesize noisy images. We evaluated our results on three test datasets: DIV2K with 100 images, KODAK24 with 24 images, and CBSD68 with 68 images.

3.4.3 Quantitative comparison

A quantitative comparison of results for image denoising methods is shown in Table 4. For each pair of results, the better result is highlighted in

Table 4 Quantitative results on benchmark datasets for image denoising methods. Each block compares results of each model trained with PE loss and the original method. Better results are highlighted in bold

Dataset	Metric	FFDNet	FFDNet-PE	RIDNet	RIDNet-PE	SADNet	SADNet-PE	NBNet	NBNet-PE	CTNet	CTNet-PE
DIV2K- $\sigma=25$	LPIPS ↓	0.1016	0.1004	0.0843	0.0792	0.0752	0.0700	0.0847	0.0754	0.1326	0.1246
	NIQE ↓	3.9367	3.8645	3.7129	3.5905	3.6382	3.5522	3.8978	3.7002	3.6700	3.5663
	SSIM ↑	0.8677	0.8695	0.8761	0.8754	0.8799	0.8789	0.8745	0.8759	0.9100	0.9099
	PSNR ↑	31.73	31.45	32.08	31.64	32.22	31.78	31.98	31.68	33.39	33.35
DIV2K- $\sigma=50$	LPIPS ↓	0.2135	0.2024	0.1525	0.1458	0.1389	0.1338	0.1491	0.1350	0.2062	0.2035
	NIQE ↓	4.9232	4.7723	4.0088	3.8035	4.0928	3.9242	4.4062	4.1284	3.9226	3.9194
	SSIM ↑	0.7704	0.7772	0.7913	0.7893	0.7973	0.7956	0.7905	0.7921	0.8490	0.8497
	PSNR ↑	26.85	26.89	27.21	26.95	27.30	27.02	27.16	26.98	30.22	30.20
DIV2K- $\sigma=75$	LPIPS ↓	0.3202	0.3096	0.2113	0.2010	0.1962	0.1901	0.2003	0.1844	0.2633	0.2589
	NIQE ↓	6.2382	6.0607	4.3089	3.9797	4.5661	4.3459	4.7297	4.4178	4.0626	4.2912
	SSIM ↑	0.6794	0.6865	0.7232	0.7231	0.7287	0.7292	0.7229	0.7253	0.8016	0.8020
	PSNR ↑	23.16	23.26	23.66	23.59	23.70	23.59	23.62	23.55	28.42	28.39
KODAK- $\sigma=25$	LPIPS ↓	0.1008	0.0925	0.0837	0.0756	0.0746	0.0686	0.0839	0.0730	0.1247	0.1124
	NIQE ↓	3.6738	3.7038	3.6222	3.6347	3.6311	3.6268	3.6911	3.6613	3.5545	3.4576
	SSIM ↑	0.8724	0.8749	0.8797	0.8804	0.8838	0.8838	0.8774	0.8800	0.8902	0.8902
	PSNR ↑	31.50	31.19	31.93	31.49	32.08	31.62	31.79	31.46	32.72	32.69
KODAK- $\sigma=50$	LPIPS ↓	0.2360	0.2198	0.1665	0.1509	0.1499	0.1377	0.1589	0.1389	0.2047	0.2132
	NIQE ↓	5.2552	5.1572	4.1250	3.9951	4.1083	3.9996	4.2193	4.1172	3.6971	3.6604
	SSIM ↑	0.7630	0.7727	0.7882	0.7898	0.7963	0.7980	0.7873	0.7917	0.8115	0.8119
	PSNR ↑	27.35	27.45	27.84	27.62	27.99	27.73	27.75	27.59	29.55	29.56
KODAK- $\sigma=75$	LPIPS ↓	0.3689	0.3555	0.2399	0.2174	0.2252	0.2088	0.2236	0.1977	0.2874	0.2802
	NIQE ↓	7.1865	7.0795	4.7059	4.4841	4.8343	4.5399	4.8031	4.7203	3.9722	4.2865
	SSIM ↑	0.6675	0.6763	0.7191	0.7237	0.7255	0.7306	0.7193	0.7253	0.7530	0.7533
	PSNR ↑	24.07	24.24	24.79	24.78	24.88	24.83	24.73	24.70	27.82	27.81
CBSD- $\sigma=25$	LPIPS ↓	0.0952	0.0868	0.0754	0.0686	0.0679	0.0626	0.0756	0.0669	0.1074	0.1012
	NIQE ↓	4.2917	4.1052	4.1114	3.8838	4.1482	3.9684	4.2599	4.0700	3.8026	3.9499
	SSIM ↑	0.8695	0.8724	0.8772	0.8769	0.8812	0.8807	0.8761	0.8777	0.8933	0.8935
	PSNR ↑	30.45	30.04	30.74	30.18	30.87	30.28	30.70	30.22	31.53	31.50
CBSD- $\sigma=50$	LPIPS ↓	0.2387	0.2191	0.1640	0.1480	0.1486	0.1365	0.1580	0.1386	0.2073	0.2038
	NIQE ↓	5.3695	5.2945	4.4370	4.2751	4.7988	4.5846	4.9306	4.7111	4.2612	4.1112
	SSIM ↑	0.7432	0.7565	0.7701	0.7718	0.7762	0.7776	0.7696	0.7747	0.8048	0.8016
	PSNR ↑	26.23	26.29	26.61	26.27	26.70	26.32	26.58	26.31	28.25	28.23
CBSD- $\sigma=75$	LPIPS ↓	0.3776	0.3602	0.2405	0.2160	0.2269	0.2075	0.2267	0.2000	0.2792	0.2757
	NIQE ↓	7.1174	7.1030	4.6947	4.6907	5.2424	5.3618	5.1385	5.3006	4.6225	4.2497
	SSIM ↑	0.6270	0.6397	0.6878	0.6943	0.6916	0.6974	0.6879	0.6961	0.7398	0.7394
	PSNR ↑	23.05	23.22	23.69	23.60	23.73	23.60	23.68	23.57	26.52	26.49

bold. For the FFDNet method, our PE loss improved metrics LPIPS, NIQE, and SSIM, and even PSNR on average. For RIDNet, SADNet, and CTNet methods, the metrics LPIPS and NIQE are better with PE loss, SSIM matches that of the original method, and PSNR is better for the original method because L_1 loss directly optimizes PSNR. For the NBNet method, all metrics except PSNR are improved with PE loss. We can see from these quantitative testing results that methods with PE loss again achieve higher perceptual quality on the image denoising task.

3.4.4 Qualitative comparison

Figure 9 compares visual effects of state-of-the-art methods FFDNet, SADNet, and NBNet for image denoising, with and without PE loss. We compare results for the DIV2K dataset, with a noise level of $\sigma = 25$. We can see from this figure that all methods with PE loss preserve the structure of images better and retain more texture details, demonstrating the advantages of PE loss in image denoising.

3.5 PE loss and image deblurring

3.5.1 Backbone network architectures

For image deblurring, we evaluated PE loss on state-of-the-art network architectures: MPRNet [39], NAFNet, HINet, Restormer, and LoFormer. MPRNet uses Charbonnier loss and edge loss, while NAFNet and HINet use PSNR loss, and Restormer uses L_1

loss. The models trained with PE loss are denoted MPRNet-PE, NAFNet-PE, HINet-PE, Restormer-PE, and LoFormer-PE, respectively.

3.5.2 Datasets

All the original methods used the GoPro dataset that contains 2103 image pairs for training and 1111 pairs for evaluation. In addition, we also tested on the HIDE dataset (with 2025 images), and two RealBlur dataset: RealBlur-J contains 980 images and RealBlur-R contains 980 images for testing.

3.5.3 Quantitative comparison

Quantitative experimental results for the image deblurring task are shown in Table 5. For each pair of results, the better result is highlighted in bold. We can see that MPRNet-PE achieves better LPIPS and NIQE, and MPRNet has better SSIM and PSNR. The other networks NAFNet, HINet, Restormer, and LoFormer have the similar performance to MPRNet. These quantitative results demonstrate the effectiveness of PE loss and show that it provides better perceptual quality. In addition, the penalty factor α also has an effect on the metrics LPIPS, NIQE, SSIM, and PSNR as shown in Table 1. For different tasks, the selection of α can be adjusted to obtain better quantitative results. However, we used the same value for α in all the image restoration tasks to ensure a fair comparison.

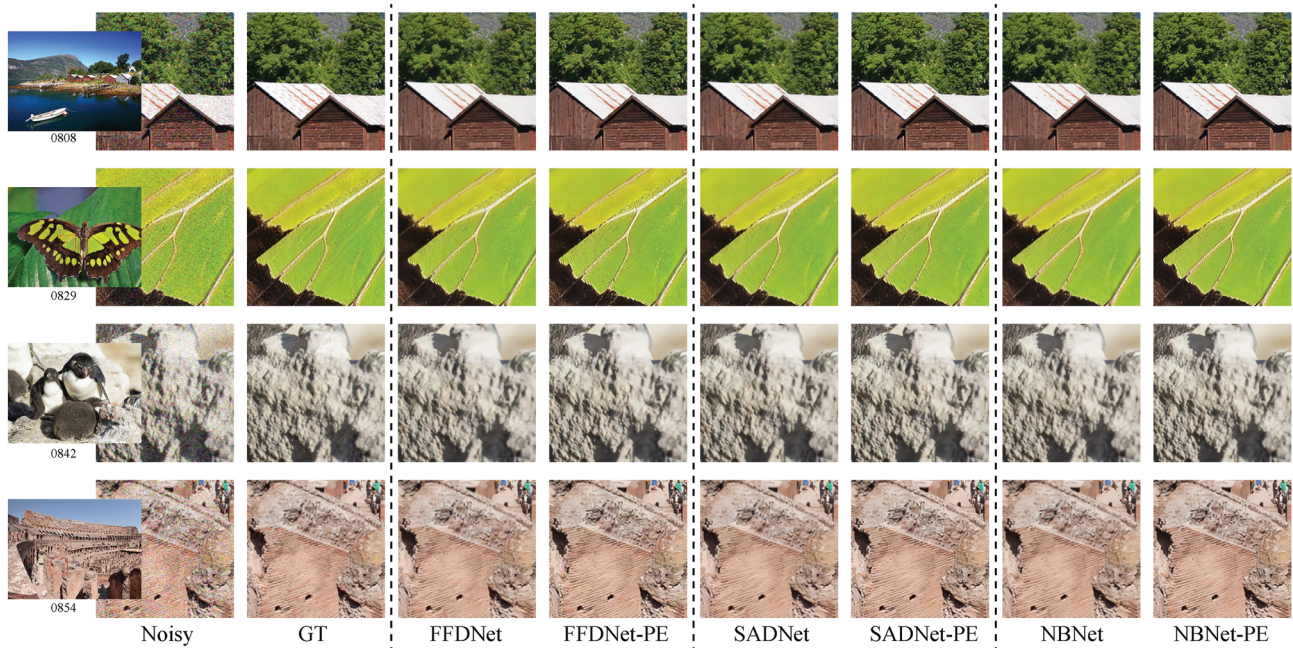


Fig. 9 Visual comparison of results of state-of-the-art network architectures for image denoising on the DIV2K dataset.

Table 5 Quantitative results on GoPro, HIDE, and RealBlur datasets for image deblurring methods. Each block compares results for each model trained with PE loss and the original method. Better results are highlighted

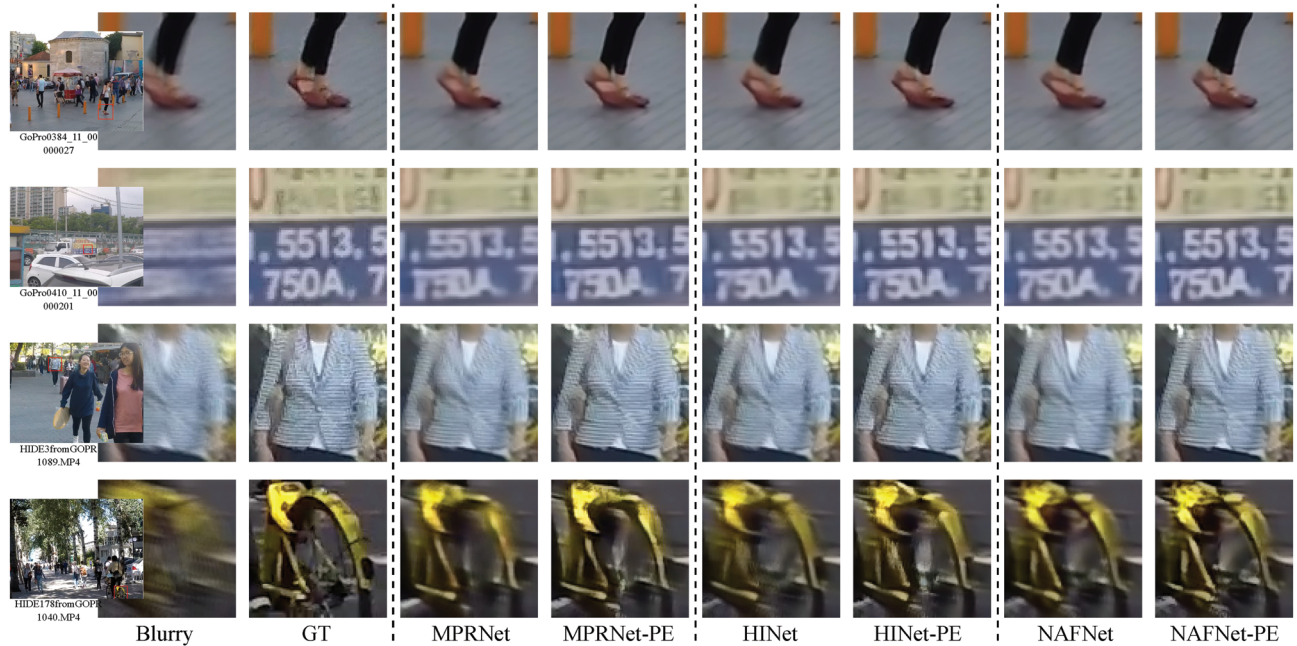
Dataset	Metric	MPRNet	MPRNet-PE	HINet	HINet-PE	NAFNet	NAFNet-PE	Restormer	Restormer-PE	LoFormer	LoFormer-PE
GoPro	LPIPS ↓	0.0606	0.0506	0.0584	0.0528	0.0546	0.0501	0.0567	0.0548	0.0759	0.0739
	NIQE ↓	5.1818	4.7570	5.1538	4.8204	5.1207	4.8078	5.2048	4.7982	5.0170	5.0101
	SSIM ↑	0.9368	0.9363	0.9387	0.9351	0.9425	0.9385	0.9404	0.9322	0.9666	0.9658
	PSNR ↑	32.67	32.36	32.91	32.31	33.09	32.46	32.93	32.06	33.79	33.70
HIDE	LPIPS ↓	0.0798	0.0637	0.0808	0.0716	0.0706	0.0624	0.0750	0.0631	0.0959	0.0934
	NIQE ↓	4.6939	4.3914	4.4491	4.3889	4.3754	4.3224	4.6852	4.4309	4.3809	4.2956
	SSIM ↑	0.9194	0.9164	0.9144	0.9084	0.9243	0.9186	0.9231	0.9201	0.9469	0.9467
	PSNR ↑	30.97	30.53	30.59	29.97	31.23	30.56	31.23	30.73	31.72	31.66
RealBlur-J	LPIPS ↓	0.1271	0.1156	0.1343	0.1341	0.1331	0.1317	0.1242	0.1136	0.1498	0.1418
	NIQE ↓	5.3445	5.0694	5.1103	4.9583	5.1016	4.9544	5.2448	5.0916	5.0827	5.0880
	SSIM ↑	0.8249	0.8072	0.8068	0.8015	0.8088	0.8002	0.8279	0.8145	0.8673	0.8665
	PSNR ↑	26.52	26.05	26.41	26.28	26.44	26.21	26.63	26.45	26.55	26.56
RealBlur-R	LPIPS ↓	0.0573	0.0522	0.0545	0.0535	0.0457	0.0453	0.0459	0.0423	0.0653	0.0624
	NIQE ↓	8.3552	8.1244	8.1411	7.9191	8.0692	7.8423	7.9832	7.8746	7.8564	7.8102
	SSIM ↑	0.9436	0.9392	0.9415	0.9398	0.9426	0.9410	0.9468	0.9446	0.9505	0.9502
	PSNR ↑	33.94	33.68	33.88	33.73	33.79	33.61	34.00	33.84	34.97	34.98

3.5.4 Qualitative comparison

Figure 10 compares the visual effects of state-of-the-art methods on the image deblurring task for MPRNet, HINet, and NAFNet, with and without PE loss, on GoPro and HIDE datasets. We can see that all the methods using PE loss obtain considerable gains over the original method, with clearer texture structures and details, demonstrating the effectiveness of PE loss for image deblurring.

3.6 User study

To assess whether the proposed PE loss improves results according to the human visual system, we conducted a subjective evaluation via a user study. In the super-resolution task, we tested the performance of ESRGAN, SPSR, EDSR, and ATD on datasets Set5, Set14, B100, and Urban100. For the denoising task, we evaluated FFDNet, SADNet, NBNet, and CTNet on DIV2K, KODAK, and CBSD datasets.

**Fig. 10** Visual comparison of the state-of-the-art network architectures for image deblurring, on GoPro and HIDE datasets.

For the deblurring task, we examined MPRNet, HINet, NAFNet, and LoFormer on GoPro, HIDE, and RealBlur datasets.

We invited 23 volunteers to participate. For each volunteer, we randomly selected 20 pairs of images to compare for each method. Volunteers were asked to vote for the images they found more visually like the ground-truth images. To mitigate random guessing in cases of similar quality, we included a “cannot decide” option. The images were presented to the voters in a random order; they used the FastStone Image Viewer for viewing and comparison. Each method received a total of 460 votes, with results summarized in Fig. 11. It is evident that methods using PE loss received more votes than original methods in all cases, indicating that the proposed PE loss can enhance subjective visual quality as judged by the human visual system. Furthermore, when considering different tasks, the denoising task received more “cannot decide” votes than in super-resolution and deblurring tasks, suggesting that some comparisons were difficult, and that the denoising task is more challenging.

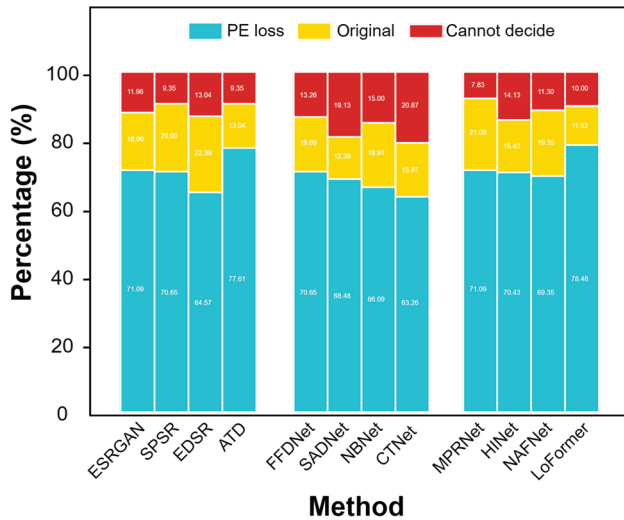


Fig. 11 User study results across different methods.

4 Conclusions

In this paper, we introduced a novel PE loss for image restoration tasks. The design of PE loss was inspired by the Mach band illusion of the human visual system. The weights of blurred pixel errors are set greater than those of other pixels. The proposed loss function detects and penalizes blurred pixels in

reconstructed images, which enhances the perceptual quality of images generated by the trained network. We use the second-order gradient map of the target image combined with the sign of the pixel difference between the reconstructed image and the target image to detect blurred pixels. Then, we penalize blurred pixels by increasing the weights of blurred pixel errors in the loss function. Experimental results on state-of-the-art methods for image super-resolution, image denoising, and image deblurring tasks demonstrate the effectiveness of our PE loss in enhancing image perceptual quality.

Appendix

Visual results from ATD [15] for the image super resolution task are shown in Fig. 12. Visual results from CTNet [22] for the image denoising task are shown in Fig. 13. Visual experimental results from LoFormer [26] for the image deblurring task are shown in Fig. 14.

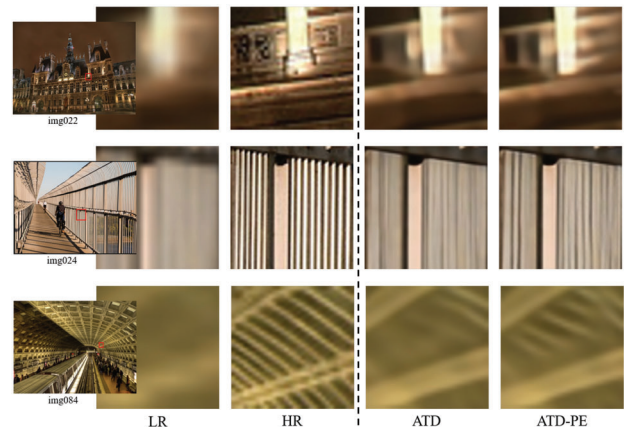


Fig. 12 Results from the ATD [15] network.

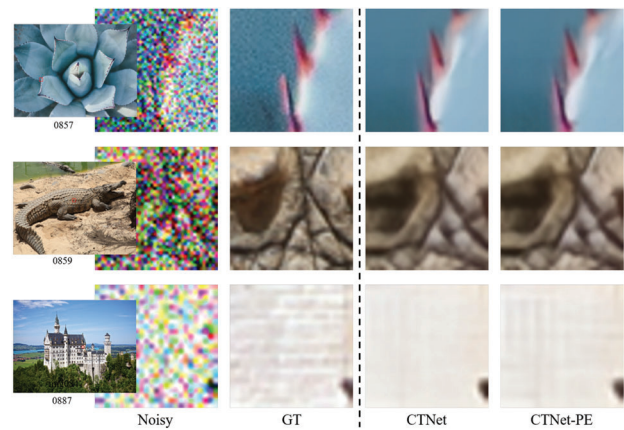


Fig. 13 Results from the CTNet [22] network.

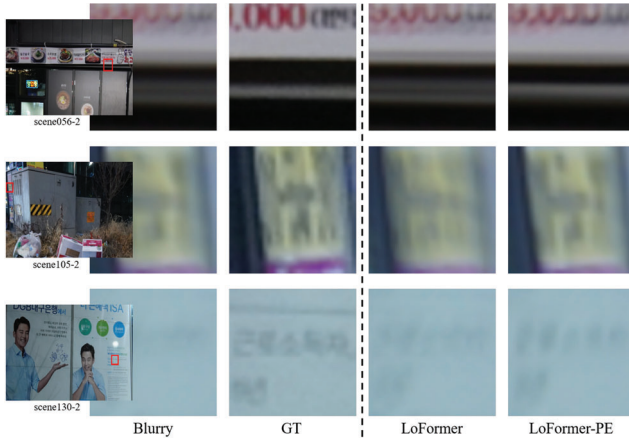


Fig. 14 Results from the LoFormer [26] network.

Numerical results from the DualCNN [40] network for image super resolution and image denoising tasks are shown in Tables 6 and 7.

Table 6 Quantitative results for the DualCNN [40] network on the image super resolution task

Dataset	Metric	DualCNN [40]	DualCNN-PE
Set5	LPIPS ↓	0.02901	0.2924
	NIQE ↓	5.5963	5.5953
	SSIM ↑	0.8186	0.8152
	PSNR ↑	28.45	28.31
Set14	LPIPS ↓	0.3868	0.3862
	NIQE ↓	4.9987	4.9957
	SSIM ↑	0.7022	0.6993
	PSNR ↑	26.22	26.19
B100	LPIPS ↓	0.4503	0.4496
	NIQE ↓	3.3536	3.3526
	SSIM ↑	0.6574	0.6551
	PSNR ↑	25.85	25.79
Urban	LPIPS ↓	0.3775	0.3772
	NIQE ↓	4.1832	4.1812
	SSIM ↑	0.6692	0.6659
	PSNR ↑	23.33	23.25

Table 7 Quantitative results for the DualCNN [40] network on the image denoising task with $\sigma=50$

Dataset	Metric	DualCNN [40]	DualCNN-PE
CBSD68	LPIPS ↓	0.2781	0.2768
	NIQE ↓	3.2676	3.2673
	SSIM ↑	0.7137	0.6942
	PSNR ↑	26.16	25.64

Author contributions

Meng Li: theoretical development, experimental design, prototype development, analysis and interpretation of data, writing the manuscript.

Yuhao Wang: experimental design, experimental setup and execution, generating figures. Yu Zhong: experimental execution, result organization, data analysis. Xishan Zhang: conception and design of work, manuscript revision. Shaoli Liu: conception and design of work, manuscript revision.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

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Meng Li is a postdoctoral researcher in the Institute of Computing Technology, Chinese Academy of Sciences. She obtained her Ph.D. degree from Peking University. Her current research focuses on image processing, machine learning, and high performance computing.

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